

توثيق مشروع | Software Project Documentation | البرمجيات

قالب توثيق مشاريع البرمجيات — للمسابقة الأفريقية الآسيوية للتكنولوجيا

1. Project Overview | نظرة عامة على المشروع

Field	Value
Project Title	NEO Challenge Database
Project Type	Web App
Description	A web application that helps users find and participate in various challenges including sports, language learning, and general knowledge. It provides personalized nutrition plans based on BMI and fitness goals.
Target Users	Students, fitness enthusiasts, language learners, general public
Resolution / Platform	Desktop + Mobile (responsive)
Date	2026-08-09

Tech Stack | Technology Stack

Category	Technology	Purpose
Language	Python	Backend logic and web interface
Framework	Streamlit	Web framework
Database	SQLite	Store user data, challenges, activities, nutrition plans

Dependencies | المكتبات المستخدمة

Package	Version	Purpose
streamlit	latest	Web framework
sqlite3	built-in	Database

2. Problem Statement | بيان المشكلة

هنا تكتب المشكلة التي يحلها مشروعك — لماذا هذا المشروع مهم؟

ما المشكلة التي يحلها هذا المشروع؟ | What problem does this project solve?

الناس يبحثون عن مواقع التدريب والكورسات والدورات التدريبية ويتطلبون وقتاً طويلاً للبحث عنها.

لماذا هذا مهم؟ | Why does it matter?

لأنها توفر وقت لاغلب الناس وتساعدهم في التغيير بعض العادات للناس بشكل أفضل.

كيف تُحل المشكلة حالياً؟ | How is it currently solved?

الموقع يحل مشكله الوقت عند معظم الناس, لأن اغلب الناس يبحثون عن مواقع التدريب والكورسات والدورات التدريبية ويتطلبون وقتاً طويلاً للبحث عنها.

3. Technical Architecture | البنية التقنية

هنا تشرح كيف مشروعك مبني من الداخل — هيكل الملفات والتقنيات المستخدمة

هيكل المشروع | Project Structure

```
plaintext
neo_challenge_database/
├── app.py                # Main Streamlit app
├── pages/                # Streamlit pages
│   ├── login.py         # User login page
│   ├── register.py      # User registration page
│   ├── view_challenges.py # View all challenges
│   ├── view_activities.py # View activities in challenges
│   ├── view_nutration_plan.py # View nutrition plans
│   ├── create_catogary.py # Create challenge categories
│   ├── create_challenges.py # Create new challenges
│   ├── create_nutration_plan.py # Create nutrition plans
│   └── view_users.py    # View all users
├── database/            # Database files
│   ├── database.db      # SQLite database
│   ├── tables.sql       # Database schema
│   ├── insert.sql       # Insert statements
│   ├── insert_challenges.sql
│   └── insert_nutrition_plan.sql
```

```
| └─ insert_users.sql
└─ docs/                # Documentation
    └─ afro_asian_software_template.md
```

Architecture Diagram | مخطط البنية

```
plaintext
[User] → [Streamlit Browser] → [Python Backend] → [SQLite Database]
```

Key Files | الملفات الرئيسية

File	Purpose	Lines of Code
app.py	Main app entry point	~1
pages/login.py	User authentication	~21
pages/register.py	User registration	~30
pages/view_challenges.py	Display all challenges	~40
database/tables.sql	Database schema	~164

4. Features & User Flow | الميزات وتدفق المستخدم

هنا تكتب قائمة الميزات + مخطط تدفق المستخدم عبر الشاشات

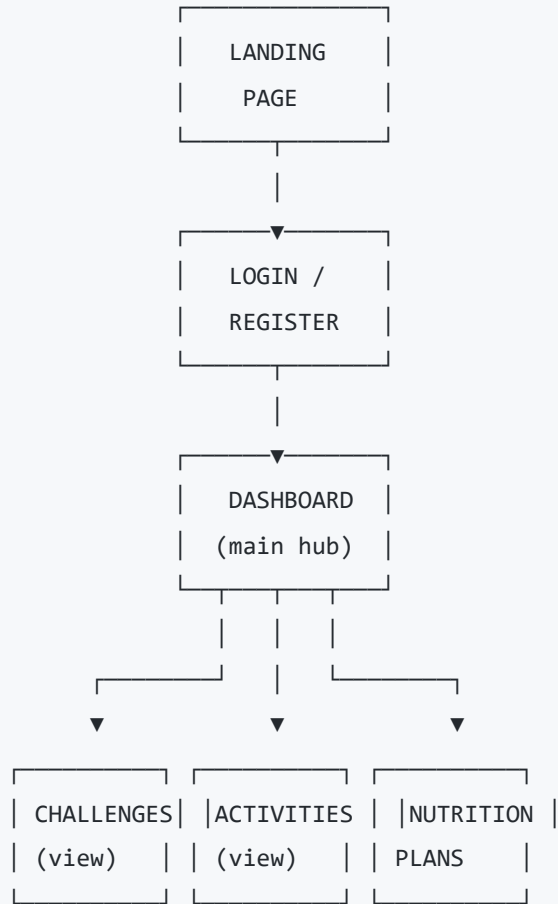
Feature List | قائمة الميزات

#	Feature	Description	Status
1	بيسجل حساب	يسجل البيانات بتاعت المستخدم	✓
2	تسجيل الدخول	يسمح للمستخدمين بالدخول لحساباتهم	✓
3	عرض التحديات	يعرض جميع التحديات المتاحة (رياضة، لغات، معرفة عامة)	✓
4	عرض الأنشطة	يعرض الأنشطة داخل كل تحدي	✓
5	عرض خطط التغذية	والهدف BMI يعرض خطط التغذية حسب	✓

Status: ✓ Complete | 🕒 In Progress | ✗ Not Started

User Flow | تدفق المستخدم

plaintext



[Replace with your actual user flow]

Screen Reference | دليل الشاشات

Screen	File / Route	Description	Transitions To
Landing	app.py	Welcome page with login/register	Register, Login
Register	pages/register.py	Create new account with user data	Login
Login	pages/login.py	User authentication	Dashboard
View Challenges	pages/view_challenges.py	Display all available challenges	View Activities
View Activities	pages/view_activities.py	Show activities within a challenge	Dashboard
View Nutrition Plans	pages/view_nutration_plan.py	Display nutrition plans based on BMI	Dashboard
Create Category	pages/create_catogary.py	Admin page to create challenge categories	Dashboard
Create Challenge	pages/create_challenges.py	Admin page to create new challenges	Dashboard
Create Nutrition Plan	pages/create_nutration_plan.py	Admin page to create nutrition plans	Dashboard

تصميم واجهة المستخدم | 5. UI/UX Design

هنا تحط لقطات الشاشة + وصف التصميم البصري لمشروعك

لقطات الشاشة | Screenshots

Landing Page: [describe what the user sees]

Dashboard: [describe what the user sees]

Key Feature: [describe what the user sees]

نظام التصميم | Design System

Element	Style
Primary Color	[e.g., #1a73e8 (blue)]
Secondary Color	[e.g., #34a853 (green)]
Font	[e.g., Inter, Arial, Cairo]
Button Style	[e.g., rounded, flat, gradient]
Layout	[e.g., sidebar navigation, top navbar, cards]

التصميم المتجاوب | Responsive Design

Breakpoint	Layout	Tested?
Desktop (1024px+)	[describe layout]	☐
Tablet (768-1024px)	[describe layout]	☐
Mobile (< 768px)	[describe layout]	☐

6. Data Sources | مصادر البيانات

ملفات، أو داتا مكتوبة في الكود، API، هنا تكتب منين جبت الداتا — قاعدة بيانات

مصادر البيانات | Data Sources

Source	Type	Description
database.db	SQLite Database	Stores user accounts, challenges, activities, nutrition plans
User Input	Form Submissions	Registration data, challenge creation, nutrition plan creation
Streamlit Session	Session State	Current logged-in user

مخطط قاعدة البيانات | Database Schema

Table	Columns	Description
users	id, username, email, password, DOB, height, weight, phone, ssn, created_at	User accounts with personal data
challenges	id, name, number_of_levels, number_of_activities, category, duration, description, created_at	Available challenges (sports, language, general knowledge)
activities	id, name, description, category, difficulty, challenge_id, created_at	Activities within each challenge
user_challenges	user_id, challenge_id	Many-to-many relationship between users and challenges
Nutrition_Plans	id, bmi, category, goal, calories, breakfast, lunch, dinner, water_intake, created_at	Nutrition plans based on BMI
User_Nutrition_Plans	user_id, nutrition_plan_id	Relationship between users and nutrition plans

واجهات برمجة التطبيقات | External APIs (if any)

API	Purpose	Rate Limit
None	All data is local	N/A

المحتوى التعليمي | 7. Educational Content

هنا تكتب المحتوى التعليمي — ماذا يتعلم الطالب من هذا المشروع؟ كيف يتعلق بالمنهج؟

أهداف التعلم | Learning Objectives

By the end of this project, the student will be able to:

1. Build a full-stack web application using Python and Streamlit
2. Design and implement a SQLite database with multiple tables and relationships
3. Implement user registration and authentication systems
4. Create and manage challenge-based learning content
5. Develop personalized nutrition planning based on BMI calculations

السياق الثقافي / التاريخي | Cultural / Historical Context

Topic	Description
Health & Fitness	The project promotes healthy lifestyles through structured challenges and nutrition planning
Language Learning	Supports multilingual education with language learning challenges
Personalized Learning	Adapts to individual user needs through BMI-based nutrition plans

التوافق مع المنهج | Curriculum Alignment

Skill Area	What the Student Practices
Problem Solving	Identifying user needs for challenges and nutrition planning
Computational Thinking	Database design, data modeling, user authentication logic
Creativity	Designing challenge categories and nutrition plan structures
Collaboration	Working with team members on different features
Technical Writing	Documenting architecture and writing this template

خطة تسويقية | 8. Marketing Plan

Short version — for full marketing plans, use the main Afro-Asian template.

نسخة مختصرة — للخطط التسويقية الكاملة، استخدم القالب الرئيسي

الجمهور المستهدف | Target Audience

- Primary:** Students ages 12-25 who want to participate in fitness, language, and knowledge challenges
- Secondary:** Teachers and educators looking for challenge-based learning tools
- Tertiary:** Fitness enthusiasts and language learners

المنصات | Platforms

- Web browser (Chrome, Firefox, Edge)
- Mobile phone (responsive design)
- School computers

Promotion Ideas | أفكار ترويجية

1. Share demo video on YouTube/TikTok showing challenge features
2. Present at school science fair or coding competition
3. Post screenshots on class WhatsApp/Telegram groups
4. Submit to Afro-Asian Tech Forum competition

9. Student Worksheet | ورقة عمل الطالب

Instructions: Copy this section into your own document and fill it in for YOUR project.

My Project Information | معلومات مشروعي

Student Name: _____

Project Title: _____

Date: _____

What is your project about? | عن ماذا يتحدث مشروعي؟

What problem does it solve? | ما المشكلة التي يحلها؟

My Tech Stack | التقنيات المستخدمة

My project type (circle one): Web App / Desktop App / Mobile App / Python Script / Other:

Technologies I used:

Category	Technology
Language	
Framework	
Database	
Libraries	

My Screens | شاشات مشروعك

Draw or describe all the screens in your project:

Screen Name	What the user sees/does	How to get here

My Features | ميزات مشروعك

Feature	What it does	Status
		✓ / ⏳ / ✗
		✓ / ⏳ / ✗
		✓ / ⏳ / ✗

My Data | بيانات مشروعك

Where does data come from?

What data is stored?

What I Learned | ما تعلمته

1. The hardest part of building my project was: _____
2. The most fun part of building my project was: _____
3. If I had more time, I would add: _____
4. One thing I would do differently: _____

Screenshot of My Project | لقطة شاشة من مشروعك

Instructor Notes | ملاحظات المدرس:

Techno Kids, Techno Future — Software Documentation Template

Afro-Asian Tech Forum · Software Category · 2026